



A note on the Cayley-Menger determinant and the Molecular Distance Geometry Problem

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ABSTRACT

Using information from protein geometry and distance data provided by Nuclear Magnetic Resonance (NMR) experiments, the Molecular Distance Geometry Problem (MDGP) can be solved by a combinatorial approach, called Branch-and-Prune (BP). The *primal* version of BP algorithm seeks MDGP graph realizations, while the *dual* BP looks for completions of associated partial distance matrices. These two algorithms are very similar when distance values are precise. In the literature, there are some proposals for extending the primal BP to take care of NMR uncertainties. Using Cayley-Menger determinant, we present a global optimization approach that also allows the dual BP to deal with NMR uncertainties.

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1. Introduction

The origin of Distance Geometry (DG) [25] is associated to Karl Menger, when he characterized geometric ideas using the concept of distance [26], and one of the most important applications of DG is in computational biology, where the 3D protein structure is calculated using distance data provided by Nuclear Magnetic Resonance (NMR) experiments [5].

The problem is called the *Molecular Distance Geometry Problem* (MDGP) [24], formally defined in terms of a graph $G = (V, E, d)$, where the sets V, E represent the atoms and the atomic pairs for which a distance is available, respectively (we say that G is *simple* if $u \neq v$, for $\{u, v\} \in E$, and *undirected* when $\{u, v\} = \{v, u\}$):

Definition 1. Given a simple undirected graph $G = (V, E, d)$ whose edges are weighted by $d : E \rightarrow (0, \infty)$, find a function $x : V \rightarrow \mathbb{R}^3$ such that

$$\forall \{u, v\} \in E, \|x_u - x_v\|_2 = d_{u,v}, \quad (1)$$

where $x_u = x(u)$, $x_v = x(v)$, $d_{u,v} = d(\{u, v\})$.

The Euclidean norm and the 3D space are the most common in molecular geometry, but for other applications, different norms and spaces $\mathbb{R}^k$ ($k \neq 3$) can be used [3,20,21,28].

The MDGP can be solved iteratively using a combinatorial method, called *Branch-and-Prune* (BP) [6,22], based on a vertex order [7,17,19] $v_1, \dots, v_n \in V$ such that (we denote x_i instead of x_{v_i} and d_{ij} instead of $d_{v_i v_j}$):

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1. For v_1, v_2, v_3 , there exist $x_1, x_2, x_3 \in \mathbb{R}^3$ satisfying Eqs. (1);
2. For $i > 3$, there exist $v_{i-3}, v_{i-2}, v_{i-1}$ such that

$$\{v_{i-3}, v_i\}, \{v_{i-2}, v_i\}, \{v_{i-1}, v_i\} \in E; \quad (2)$$

3. For $i > 3$,

$$d_{i-3,i-2} + d_{i-2,i-1} > d_{i-3,i-1}. \quad (3)$$

MDGP instances with this order are called the *Discretizable Molecular Distance Geometry Problem* (DMDGP) [14,15]. Property 1 says that we can start with fixed positions for v_1, v_2, v_3 . Property 2 allows us to define a system

$$\begin{aligned} \|x_i - x_{i-1}\|_2 &= d_{i-1,i}, \\ \|x_i - x_{i-2}\|_2 &= d_{i-2,i}, \\ \|x_i - x_{i-3}\|_2 &= d_{i-3,i}, \end{aligned} \quad (4)$$

which provides up to two possible positions for $v_i, i > 3$ (from Property 3), given $x_{i-1}, x_{i-2}, x_{i-3} \in \mathbb{R}^3$. For each position for v_i , we obtain another two for v_{i+1} , implying that the DMDGP search space can be represented by a binary tree [15].

For $i > 4$, we have to check if there is an additional edge $\{v_j, v_i\} \in E, j < i - 3$, implying another equation to the system related to v_i : $\|x_i - x_j\|_2 = d_{j,i}$ (such value $d_{j,i}$ is called a *pruning distance*). If the points $x_j, x_{i-1}, x_{i-2}, x_{i-3} \in \mathbb{R}^3$ are not in the same plane, we obtain a unique solution x_i^* for v_i , supposing $\|x_i^* - x_j\|_2 = d_{j,i}$. However, both positions for v_i may not satisfy the additional equation and must be pruned. In that case, we consider the other possible positions for v_{i-1} and repeat the procedure [15]. The search ends when a path from the root of the tree to a leaf node is found, in such a way that the positions for all the vertices in the DMDGP order satisfy Eqs. (1).

Distances $d_{i-1,i}$ and $d_{i-2,i}$ are known *a priori*, considered as precise values, since they are related to bond lengths and bond angles of a protein. However, if distances $d_{i-3,i}$ are provided by NMR experiments, they contain uncertainties [4].

In [16], distances $d_{i-3,i}$ were represented as interval distances $[\underline{d}_{i-3,i}, \bar{d}_{i-3,i}]$, where $\underline{d}_{i-3,i} \leq d_{i-3,i} \leq \bar{d}_{i-3,i}$, and an extension of the BP algorithm was proposed, called *iBP*, where the idea is to sample values from $[\underline{d}_{i-3,i}, \bar{d}_{i-3,i}]$ in order to solve the system (4) related to v_i [10,12]. Choosing many values, the search space increases exponentially, and for small samples, a solution may not be found [7,30].

For the *branching phase* of *iBP*, there are two recent and different approaches to avoid such sampling strategy: one is based on geometric algebra [1,2] and the other uses single-level interval analysis [9]. For the *pruning phase*, however, there is not a well established solution method. Some results are given in [11,18].

All strategies mentioned above for dealing with NMR uncertainties are given in terms of the atomic Cartesian coordinates. In [23], a *dual BP* algorithm is proposed to solve DMDGP instances based only on the calculation of (precise) distance values. Since the NMR data are provided as interval distances, the dual BP must be able to consider this kind of data. This is our motivation in this paper.

The next section explains all the details of our contribution and Section 3 provides some computational results. We conclude the paper with new research directions.

2. Cayley-Menger Determinant and the DMDGP

In the definition of the DMDGP, the strict triangular inequality (Property 3) is required to guarantee a finite search space, and since the bond angles in protein molecules are not linear, this is a realistic assumption [5].

The triangular inequality can be generalized using the so called Cayley-Menger determinant [8,27], defined for a set $S_{n+1} = \{p_0, p_1, \dots, p_n\}$ of points in $\mathbb{R}^n$, given by

$$CM(S_{n+1}) = \begin{vmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{0,1}^2 & \dots & d_{0,n}^2 \\ 1 & d_{0,1}^2 & 0 & \dots & d_{1,n}^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & d_{0,n}^2 & d_{1,n}^2 & \dots & 0 \end{vmatrix},$$

where $d_{ij} = \|p_i - p_j\|_2, \forall i, j \in \{0, \dots, n\}$.

It is possible to prove that five points in $\mathbb{R}^3$ have zero Cayley-Menger determinant [29], and using this fact for the DMDGP order, considering $S_5 = \{x_{i-4}, x_{i-3}, x_{i-2}, x_{i-1}, x_i\}$, we get

$$CM(S_5) = 0 \Rightarrow \begin{vmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{i-4,i-3}^2 & d_{i-4,i-2}^2 & d_{i-4,i-1}^2 & \delta^2 \\ 1 & d_{i-4,i-3}^2 & 0 & d_{i-3,i-2}^2 & d_{i-3,i-1}^2 & d_{i-3,i}^2 \\ 1 & d_{i-4,i-2}^2 & d_{i-3,i-2}^2 & 0 & d_{i-2,i-1}^2 & d_{i-2,i}^2 \\ 1 & d_{i-4,i-1}^2 & d_{i-3,i-1}^2 & d_{i-2,i-1}^2 & 0 & d_{i-1,i}^2 \\ 1 & \delta^2 & d_{i-3,i}^2 & d_{i-2,i}^2 & d_{i-1,i}^2 & 0 \end{vmatrix} = 0,$$

resulting in a quadratic equation with two possible values for δ^2 , where δ is the distance associated to the pair $\{i-4, i\}$. This is the equation that must be solved, in the dual BP, replacing the *primal* Eq. (4).

When NMR uncertainties are taken into account, distance values $d_{i-4,i-1}$ and $d_{i-3,i}$ may be represented as interval distances. Instead of sampling values from intervals $[\underline{d}_{i-4,i-1}, \bar{d}_{i-4,i-1}]$ and $[\underline{d}_{i-3,i}, \bar{d}_{i-3,i}]$, we propose a new way to calculate the possible values for δ .

2.1. Calculating intervals for $d_{i-4,i}$

Rewriting $CM(S_5)$ in terms of $\underline{d}_{i-4,i-1}$ and $\underline{d}_{i-3,i}$, with $\lambda_1 \in [0, \bar{d}_{i-4,i-1} - \underline{d}_{i-4,i-1}]$ and $\lambda_2 \in [0, \bar{d}_{i-3,i} - \underline{d}_{i-3,i}]$, we have

$$\begin{vmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{i-4,i-3}^2 & d_{i-4,i-2}^2 & (\underline{d}_{i-4,i-1} + \lambda_1)^2 & \delta^2 \\ 1 & d_{i-4,i-3}^2 & 0 & d_{i-3,i-2}^2 & d_{i-3,i-1}^2 & (\underline{d}_{i-3,i} + \lambda_2)^2 \\ 1 & d_{i-4,i-2}^2 & d_{i-3,i-2}^2 & 0 & d_{i-2,i-1}^2 & d_{i-2,i}^2 \\ 1 & (\underline{d}_{i-4,i-1} + \lambda_1)^2 & d_{i-3,i-1}^2 & d_{i-2,i-1}^2 & 0 & d_{i-1,i}^2 \\ 1 & \delta^2 & (\underline{d}_{i-3,i} + \lambda_2)^2 & d_{i-2,i}^2 & d_{i-1,i}^2 & 0 \end{vmatrix} = 0.$$

Solving this equation, we obtain the following functions:

$$\begin{aligned} \delta_1^2(\lambda_1, \lambda_2) = & [c_1 + c_2\lambda_1 + c_3\lambda_1^2 + c_4\lambda_2 + c_5\lambda_1\lambda_2 \\ & + c_6\lambda_1^2\lambda_2 + c_7\lambda_2^2 + c_8\lambda_1^2\lambda_2^2 + c_9\lambda_1\lambda_2^2 \\ & + (c_{10} + c_{11}\lambda_1 + c_{12}\lambda_1^2 + c_{13}\lambda_2 + c_{14}\lambda_1\lambda_2 \\ & + c_{15}\lambda_1^2\lambda_2 + c_{16}\lambda_1^2\lambda_2^2 \\ & + c_{17}\lambda_1^3 + c_{18}\lambda_1^4 + c_{19}\lambda_1^3\lambda_2 + c_{20}\lambda_1^4\lambda_2 + c_{21}\lambda_2^2 \\ & + c_{22}\lambda_1\lambda_2^2 + c_{23}\lambda_1^3\lambda_2^2 + c_{24}\lambda_1^4\lambda_2^2 \\ & + c_{25}\lambda_2^3 + c_{26}\lambda_1\lambda_2^3 + c_{27}\lambda_1^2\lambda_2^3 + c_{28}\lambda_1^3\lambda_2^3 + c_{29}\lambda_1^4\lambda_2^3 \\ & + c_{30}\lambda_2^4 + c_{31}\lambda_1\lambda_2^4 + c_{32}\lambda_1^2\lambda_2^4 + c_{33}\lambda_1^3\lambda_2^4 + c_{34}\lambda_1^4\lambda_2^4)^{1/2}] \times c_{35} \end{aligned} \quad (5)$$

and

$$\begin{aligned} \delta_2^2(\lambda_1, \lambda_2) = & [c_1 + c_2\lambda_1 + c_3\lambda_1^2 + c_4\lambda_2 + c_5\lambda_1\lambda_2 \\ & + c_6\lambda_1^2\lambda_2 + c_7\lambda_2^2 + c_8\lambda_1^2\lambda_2^2 + c_9\lambda_1\lambda_2^2 \\ & - (c_{10} + c_{11}\lambda_1 + c_{12}\lambda_1^2 + c_{13}\lambda_2 + c_{14}\lambda_1\lambda_2 \\ & + c_{15}\lambda_1^2\lambda_2 + c_{16}\lambda_1^2\lambda_2^2 \\ & + c_{17}\lambda_1^3 + c_{18}\lambda_1^4 + c_{19}\lambda_1^3\lambda_2 + c_{20}\lambda_1^4\lambda_2 + c_{21}\lambda_2^2 \\ & + c_{22}\lambda_1\lambda_2^2 + c_{23}\lambda_1^3\lambda_2^2 + c_{24}\lambda_1^4\lambda_2^2 \\ & + c_{25}\lambda_2^3 + c_{26}\lambda_1\lambda_2^3 + c_{27}\lambda_1^2\lambda_2^3 + c_{28}\lambda_1^3\lambda_2^3 + c_{29}\lambda_1^4\lambda_2^3 \\ & + c_{30}\lambda_2^4 + c_{31}\lambda_1\lambda_2^4 + c_{32}\lambda_1^2\lambda_2^4 + c_{33}\lambda_1^3\lambda_2^4 + c_{34}\lambda_1^4\lambda_2^4)^{1/2}] \times c_{35}, \end{aligned} \quad (6)$$

where the values for c_i , $i = 1, \dots, 35$, are available at <https://cutt.ly/BfPgzoee>.

Since $\delta_1^2(\lambda_1, \lambda_2)$ and $\delta_2^2(\lambda_1, \lambda_2)$ are continuous functions, for $\lambda_1 \in [0, \bar{d}_{i-4,i-1} - \underline{d}_{i-4,i-1}]$ and $\lambda_2 \in [0, \bar{d}_{i-3,i} - \underline{d}_{i-3,i}]$, $\delta_1^2(\lambda_1, \lambda_2)$ and $\delta_2^2(\lambda_1, \lambda_2)$ result in other two intervals, implying that the possible values for $d_{i-4,i}$ ($\delta_t, t = 1, 2$) can then be calculated by solving the following optimization problems:

$$\min_{\lambda_1, \lambda_2 \in \mathbb{R}} \delta_t^2(\lambda_1, \lambda_2)$$

$$\lambda_1 \in [0, \bar{d}_{i-4,i-1} - \underline{d}_{i-4,i-1}]$$

$$\lambda_2 \in [0, \bar{d}_{i-3,i} - \underline{d}_{i-3,i}]$$
(7)

and

$$\max_{\lambda_1, \lambda_2 \in \mathbb{R}} \delta_t^2(\lambda_1, \lambda_2)$$

$$\lambda_1 \in [0, \bar{d}_{i-4,i-1} - \underline{d}_{i-4,i-1}]$$

$$\lambda_2 \in [0, \bar{d}_{i-3,i} - \underline{d}_{i-3,i}]$$
(8)

3. Computational results

Since the proposed method can be used for any sequence of five connected atoms, we will generate random DMDGP instances with that size, according to the suggestions in [13].

By fixing the lengths of covalent bonds ($d_{i-1,i} = 1.0$) and the values of covalent angles ($\theta_{i-2,i} = 2.0$ radians), instances with five atoms are defined by torsion angles $\omega_{i-4,i-1} \in [0, 2\pi]$ and $\omega_{i-3,i} \in [0, 2\pi]$. We randomly select a value ω from the set $\{60^\circ, 180^\circ, 300^\circ\}$, making a perturbation randomly selecting another value from $[0.9\omega, 1.1\omega]$.

In order to calculate the associated Cartesian coordinates of the points $(x_i, y_i, z_i) \in \mathbb{R}^3$, $i = 1, \dots, 5$, we use the following homogeneous matrices [15]:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix} = B_1 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = B_1 B_2 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} x_3 \\ y_3 \\ z_3 \\ 1 \end{bmatrix} = B_1 B_2 B_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

and

$$\begin{bmatrix} x_4 \\ y_4 \\ z_4 \\ 1 \end{bmatrix} = B_1 B_2 B_3 B_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} x_5 \\ y_5 \\ z_5 \\ 1 \end{bmatrix} = B_1 B_2 B_3 B_4 B_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

where

$$B_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_2 = \begin{bmatrix} -1 & 0 & 0 & -d_{1,2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$B_3 = \begin{bmatrix} -\cos \theta_{1,3} & -\sin \theta_{1,3} & 0 & -d_{2,3} \cos \theta_{1,3} \\ \sin \theta_{1,3} & -\cos \theta_{1,3} & 0 & d_{2,3} \sin \theta_{1,3} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and

$$B_i = \begin{bmatrix} -\cos \theta_{i-2,i} & -\sin \theta_{i-2,i} & 0 & -d_{i-1,i} \cos \theta_{i-2,i} \\ \sin \theta_{i-2,i} \cos \omega_{i-3,i} & -\cos \theta_{i-2,i} \cos \omega_{i-3,i} & -\sin \omega_{i-3,i} & d_{i-1,i} \sin \theta_{i-2,i} \cos \omega_{i-3,i} \\ \sin \theta_{i-2,i} \sin \omega_{i-3,i} & -\cos \theta_{i-2,i} \sin \omega_{i-3,i} & \cos \omega_{i-3,i} & d_{i-1,i} \sin \theta_{i-2,i} \sin \omega_{i-3,i} \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for $i \in \{4, 5\}$.

From the calculations above, we can obtain the distances between the atomic pairs $i-4, i-1$ and $i-3, i$.

We will assume that these distances are equal to $\underline{d}_{i-4,i-1}$ and $\underline{d}_{i-3,i}$, respectively, and uncertainties will be given by $\lambda_1, \lambda_2 \in \{0.05, 0.10, 0.15, 0.20\}$.

To solve all the instances, we used the software AMPL, with the solver Baron 17.4.1, on a Lenovo notebook with 6 GB RAM and Intel celeron 1.6 GHz.

All generated data and the AMPL model file are available at <https://cutt.ly/BfPgzoE>. Tables 1–4 provide the results.

4. Conclusions

The dual BP algorithm, proposed in [23], works in the *distance space*. This means that while the primal BP seeks graph realizations, the dual BP seeks completions of partial distance matrices, making realizations and distance matrix completions dual to each other.

When distances are precise values, both algorithms are similar. However, to deal with interval distances, the dual BP appears to be more suitable, since the input data of the problem are distance values. For the first time, this paper presents some new results on this research direction.

The intervals obtained for $d_{i-4,i}$ are the mathematically possible values, calculated in terms of the distances

$$d_{i-4,i-3}, d_{i-4,i-2}, d_{i-3,i-2}, d_{i-3,i-1}, d_{i-2,i-1}, d_{i-2,i}, d_{i-1,i},$$

and

$$[d_{i-3,i}, \bar{d}_{i-3,i}], [d_{i-4,i-1}, \bar{d}_{i-4,i-1}],$$

which can be very useful to validate NMR data.

Table 1

Computational time and solutions obtained for $\lambda_1 \leq 0.05$ and $\lambda_2 \leq 0.05$.

$\omega_{i-4,i-1}$	$\omega_{i-3,i}$	$\underline{d}_{i-4,i-1}$	$\underline{d}_{i-3,i}$	$d_{i-4,i}$	Time (s)
300.08	327.63	2.0449	1.9010	$[1.7675, 1.8732] \cup [2.1577, 2.3683]$	0.41
177.78	191.20	2.5813	2.5754	$[3.3584, 3.5107]$	0.25
188.12	321.31	2.5783	1.9287	$[2.8652, 2.9541]$	0.36
302.14	62.63	2.0325	2.0617	$[1.8973, 2.0202] \cup [2.4671, 2.6322]$	0.27
196.03	323.16	2.5690	1.9202	$[2.8223, 2.9615]$	0.20

Table 2

Computational time and solutions obtained for $\lambda_1 \leq 0.10$ and $\lambda_2 \leq 0.10$.

$\omega_{i-4,i-1}$	$\omega_{i-3,i}$	$\underline{d}_{i-4,i-1}$	$\underline{d}_{i-3,i}$	$d_{i-4,i}$	Time (s)
325.72	180.93	1.9090	2.5815	$[2.8798, 2.9561]$	0.28
167.72	295.05	2.5742	2.0761	$[2.9319, 3.1130]$	0.30
63.77	180.52	2.0663	2.5815	$[2.9867, 3.0640]$	0.48
171.15	163.74	2.577	2.5687	$[3.3363, 3.6520]$	0.32
183.58	302.11	2.5809	2.0326	$[2.9501, 3.0507]$	0.29

Table 3

Computational time and solutions obtained for $\lambda_1 \leq 0.15$ and $\lambda_2 \leq 0.15$.

$\omega_{i-4,i-1}$	$\omega_{i-3,i}$	$\underline{d}_{i-4,i-1}$	$\underline{d}_{i-3,i}$	$d_{i-4,i}$	Time (s)
282.06	56.52	2.1599	2.0245	$[2.0308, 2.4121] \cup [2.619, 3.0026]$	0.32
176.98	288.46	2.5811	2.1182	$[3.0109, 3.1453]$	0.26
308.00	62.73	1.9982	2.0623	$[1.8619, 2.2347] \cup [2.4048, 2.8633]$	0.37
192.77	57.69	2.5736	2.0315	$[2.9009, 3.1184]$	0.26
175.27	62.56	2.5804	2.0610	$[2.9635, 3.1101]$	0.24

Table 4

Computational time and solutions obtained for $\lambda_1 \leq 0.20$ and $\lambda_2 \leq 0.20$.

$\omega_{i-4,i-1}$	$\omega_{i-3,i}$	$\underline{d}_{i-4,i-1}$	$\underline{d}_{i-3,i}$	$d_{i-4,i}$	Time (s)
162.37	322.81	2.5664	1.9218	$[2.8139, 3.0860]$	0.24
58.16	327.80	2.0344	1.9004	$[1.7496, 2.8284]$	0.28
57.84	291.12	2.0325	2.1012	$[1.9507, 3.0548]$	0.27
307.47	197.10	2.0012	2.5673	$[2.8568, 3.1468]$	0.26
55.26	290.70	2.0171	2.1038	$[1.9394, 3.0419]$	0.24

Differently from the “classical” way to deal with interval distances, where the dual BP should sample values from the intervals $[\underline{d}_{i-3,i}, \bar{d}_{i-3,i}]$ and $[\underline{d}_{i-4,i-1}, \bar{d}_{i-4,i-1}]$ in order to calculate $d_{i-4,i}$, the new approach proposed here avoids sampling process and allows us to obtain all the possible values for $d_{i-4,i}$.

If an interval distance $d_{i-4,i}$ is provided by NMR data and the sampled value is outside this interval, another samples must be selected from $[\underline{d}_{i-3,i}, \bar{d}_{i-3,i}]$ and $[\underline{d}_{i-4,i-1}, \bar{d}_{i-4,i-1}]$. Using the proposed approach, we just intersect all the related intervals. For example, considering the last line of Table 1 and supposing that NMR provides $d_{i-4,i} \in [2.91, 3.55]$, the new interval for $d_{i-4,i}$ would be

$$d_{i-4,i} \in [2.8223, 2.9615] \cap [2.91, 3.55] = [2.91, 2.96].$$

Computational experiments indicate that torsion angle values are related to the fact that distance values for $d_{i-4,i}$ can be given by an interval or an union of intervals. Our further study will investigate this in a more detailed way.

CRedit authorship contribution statement

Luiz Leduino de Salles Neto: Investigation, Software, Data curation. **Carlile Lavor:** Methodology, Validation, Writing - review & editing. **Weldon Lodwick:** Conceptualization, Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ins.2020.12.072>.

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